

Intelligent Agents

Assignment: Designing and Evaluating an Intelligent Agent

Chosen Agent: Smart Irrigation Agent

This assignment presents the design and evaluation of an original intelligent agent — the **Smart Irrigation Agent** — covering problem identification, problem formulation, PEAS-based evaluation, and classification of its environmental properties, in line with the standard framework for analyzing intelligent agents.

1. Identify the Problem

Farmers traditionally water their crops manually, either on a fixed schedule or based on personal judgement of when the soil needs water. This informal approach leads to two recurring problems:

- **Overwatering** — wastes water and can damage crop roots.
- **Underwatering** — crops do not receive sufficient water, reducing yield.

There is currently no automatic system that decides, in real time, whether a field genuinely needs water based on actual conditions such as soil moisture, temperature, and weather.

Problem Statement

There is a need for an intelligent agent that can automatically monitor soil and weather conditions and decide when to turn irrigation (the water pump) on or off, so that crops receive the right amount of water at the right time — without wasting water or relying on manual guesswork.

2. Formulate the Problem (Text-Based)

The problem is formulated below in terms of the agent's initial state, available actions, goal state, percepts, and the sequence of actions it follows.

Initial State

The field's current condition is unknown to the system until the sensors report it — for example, the soil moisture level, temperature, and the current weather condition.

Actions Available to the Agent

- Turn the water pump ON
- Turn the water pump OFF
- Keep monitoring (take no action this cycle)

Goal State

Soil moisture stays within the ideal range for the crop (neither too dry nor too wet) — water is supplied only when it is actually needed.

Percepts (Inputs the Agent Receives)

- Soil moisture level (dry / normal / wet)

- Temperature (hot / normal / cold)
- Weather forecast (rain expected / no rain expected)

Sequence of Actions (Path)

Sense soil moisture → Check temperature → Check weather forecast → Decide (water needed or not) → Act (pump ON / OFF) → Re-check after interval → Repeat

Formulated Problem Statement

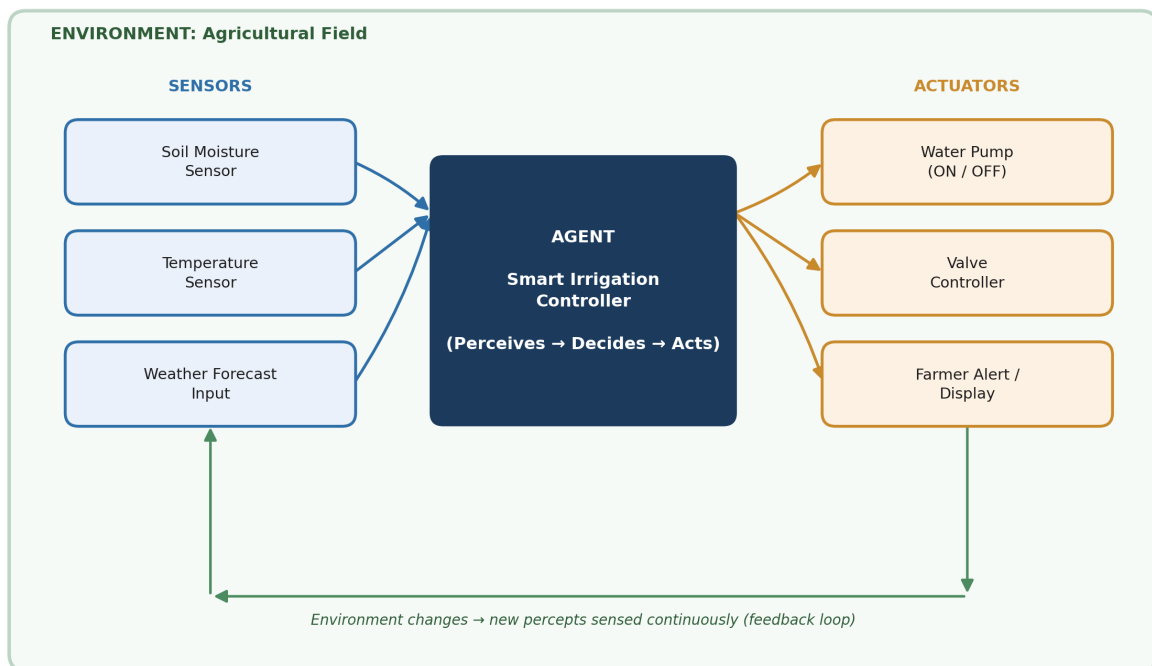
Given the current soil moisture, temperature, and weather forecast as input, the agent must decide the correct pump action (ON/OFF) at each time interval, such that soil moisture remains within the optimal range for healthy crop growth, while minimizing unnecessary water usage.

2.1 Graphical Representation of the Environment

The diagram below illustrates the agent's perception–action cycle within its environment: sensors gather information from the agricultural field, the agent processes this information and makes a decision, and actuators act back upon the environment — which in turn produces new conditions for the sensors to perceive.

Graphical Environment Representation

Smart Irrigation Agent — Perception–Action Cycle



3. Evaluate in the Context of PEAS

PEAS defines the task environment of an agent through four components: **P**erformance measure, **E**nvironment, **A**ctuators, and **S**ensors.

Component	Details
P — Performance Measure	Maintains optimal soil moisture; minimizes water wastage; reduces manual farmer effort; improves crop yield and health over time.
E — Environment	Agricultural field/farmland, soil, crops, weather conditions, and the water supply system.
A — Actuators	Water pump (ON/OFF), valve controller (opens/closes water flow to field sections), and an alert/display system that notifies the farmer of pump status.
S — Sensors	Soil moisture sensor, temperature sensor, and weather forecast input (via internet/API or local sensor).

4. Identify the Environmental Properties

The agent's environment is classified below using the standard set of environment properties.

Property	Type	Justification
Observability	Partially Observable	The agent only senses soil moisture, temperature, and weather forecast — it cannot observe everything in the field (e.g., exact plant health or underground water table), so its knowledge is incomplete.
Determinism	Stochastic	Even when the pump is turned ON, the outcome is not fully predictable — unexpected rain, sudden temperature change, or equipment issues can affect the result.
Episodes	Sequential	The agent's current decision affects future states; watering too much now keeps soil moisture high later, so past actions influence what happens next.
Change	Dynamic	The environment keeps changing on its own even while the agent is deciding — weather shifts, soil dries out, and temperature rises during the day.
Values	Continuous	Soil moisture, temperature, and water flow are continuous values (e.g., 42.5% moisture, 31.2°C) rather than a small fixed set of discrete options.
Agents	Single Agent	Only one agent — the irrigation system — operates and makes decisions in this environment; no other competing or cooperating agent is involved.

End of Assignment — Smart Irrigation Agent